

Overview

Extreme heat and rapidly developing drought are among the most damaging hazards in the United States, yet their week-3 predictability remains limited because local land-atmosphere feedbacks interact with remote teleconnection patterns and global circulation anomalies. This fellowship will develop a probabilistic, physics-guided graph neural forecasting framework for 15-day prediction of CONUS daily maximum temperature anomalies and associated flash-drought risk. The project builds from my MeshFlowNet/conditional-flow-matching model, a GraphCast-style icosahedral mesh encoder-processor-decoder trained on PRISM, ERA5, teleconnection indices, topography, seasonal radiation, soil-moisture, and global circulation fields. The proposed host organization is University of Florida, Department of Geography. The proposed scientific mentors are David Keellings, University of Florida.

Intellectual Merit

The project asks: how much week-3 predictability of U.S. heat extremes is contained in global circulation, ocean-atmosphere teleconnections, soil-moisture memory, and local persistence, and can conditional flow models extract this signal with calibrated uncertainty? Three aims will: (1) extend a deterministic 15-day residual MeshFlowNet into an ensemble conditional-flow model; (2) diagnose regional and teleconnection-dependent skill using 5-fold leave-years-out hindcasts for 1981-2023; and (3) benchmark against persistence, climatology, ridge/teleconnection baselines, and operational S2S reference skill. Preliminary hindcasts over 5,332 MJJAS samples show stitched temporal anomaly correlation of 0.1016 versus 0.0735 for persistence, motivating a focused investigation of where and why learned graph dynamics add value.

Broader Impacts

Improved week-3 heat-risk guidance can support public-health planning, water resources management, energy operations, and agricultural preparedness. The fellowship will produce open, reproducible workflows; publish model diagnostics and derived hindcast products with metadata; and translate the research into teaching modules on trustworthy AI for climate extremes. I will mentor undergraduate and REU students in climate-data science, with emphasis on transparent evaluation, uncertainty, and equitable access to AI tools for environmental risk assessment.